Matching Liquor Supply to Demand: What Drives Spirits Buying in Iowa?

1. Project and Group

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Group No.: 1

Members: Darshita Yadav (240981), Raghav Yadav (240598), Rayna Manchanda (240601), Prerit Shrivastava (240593)

2. Stakeholder and Problem

- **Stakeholder:** The Iowa Alcoholic Beverages Division (ABD), the state's only spirits wholesaler, which licenses every retailer.
- **Problem:** Demand differs sharply between counties (from 1.4 to 22.7 liters per adult in 2025), but ABD does not know which county traits predict how much, and what kind of, spirits are bought.
- **Why it matters:** Stock matched to local demand avoids stock-outs and overstock across about 2,240 stores.

3. Research Question and Hypotheses

Question: Which county characteristics explain how much spirits each county buys per adult, and which kinds it buys?

Hypotheses:

- **H1:** Counties with larger populations buy more liters of spirits per adult and place more orders per person.
- **H2:** Counties with a larger Hispanic population buy a bigger share of tequila and mezcal, even after allowing for county size and income.
- **H3:** Counties with higher median household income buy a bigger share of premium-priced spirits (higher price per liter).
- **H4:** Counties with a higher share of young adults (21–34) buy a bigger share of ready-to-drink and flavored spirits.

4. Data Sources

- **Iowa Liquor Sales, 2025 (Iowa Data Hub, dataset 1262):** every store order from the state, Jan–Dec 2025, 2.49 million rows; published by the Alcohol Operations Bureau under a CC BY licence.
- **American Community Survey 5-year, 2020–2024 (US Census Bureau):** 16 demographic tables for Iowa's 99 counties.

5. Integration Plan (Single Source of Truth)

- One table with one row per sales line (2.92 million rows × 46 columns), summarised to one row per county for testing.
- **Join key:** the 5-digit county FIPS code links every sale to its county's census profile; 99.96% of rows are matched.
- **Duplicates:** 429,996 repeated rows (Aug–Oct), almost all a copy made by the download zip, are flagged and removed.
- **Missing values and units:** 621 blank counties filled from the store's county, 1,308 dropped; Nov–Dec liters (rounded down in the source) recalculated from bottles × bottle size; units are liters and US dollars.
- **Validation:** all 99 counties are present in every source; half-year totals add up to the full year; scripts rebuild everything from the raw files.

6. Methods and BADIR Workflow

- **Business question:** Which counties need more stock, and which need a different mix of spirits?
- **Analysis plan:** Choose the outcomes (liters per adult, category shares) and the county factors to test.
- **Data:** Summarise the combined sales + census table by county.
- **Insights:** Charts and correlation tests (Spearman), a regression model with controls, then K-means clustering to group similar counties.
- **Recommendations:** three changes to ABD stocking policy.
    - **Recommendation 1:** Set a stocking profile for each county group so order volume matches local demand instead of a single statewide norm.
    - **Recommendation 2:** Adjust each county's spirits mix (e.g. tequila/mezcal share) to match its demographic profile rather than a fixed category split.
    - **Recommendation 3:** Flag high-volume counties for public-health review alongside the stocking plan, since demand concentration may also signal risk.

7. Validation and Decision Value

- A hypothesis is kept only if it passes on Jan–Jun data and again on Jul–Dec data (p-value < 0.05 after correcting for testing many hypotheses). Rejected hypotheses are also reported.
- Results must still hold after removing extreme counties and after weighting counties by population.
- **Result:** ABD can plan each county's stock level and spirits mix from its demographic profile.

8. Limitations and Next Steps

- **Limitations:** results show links, not proof of cause; sales are recorded at the store, not where the buyer lives, so tourism and cross-border shopping inflate some counties (e.g. Dickinson, Clinton); county results do not describe individuals.
- **Next steps:** run the Spearman correlation and regression models on the full-year data, finalize the K-means county groups, and translate them into per-county stocking and spirits-mix recommendations for the final report.